

a. An African cycad



b. *Ginkgo biloba,* a native of China

Figure 23.19 Three groups of gymnosperms. **a.** Cycads may resemble ferns or palms, but they are cone-producing gymnosperms. **b.** A ginkgo tree has broad leaves and fleshy seeds borne at the end of stalklike megasporophylls. **c.** *Ephedra,* a type of gnetophyte, is a branched shrub. This specimen produces pollen in microsporangia. **d.** *Welwitschia,* another type of gnetophyte, produces two straplike leaves that fray and split. (a): (cycad): Hoberman Collection/Alamy Stock Photo; (cycad cone): PlazaCameraman/iStock/Getty Images; (b): (Ginkgo leaves): blickwinkel/Koenig/Alamy Stock Photo; (ovule): Ken Robertson, University of Illinois/INHS; (c): (plant): Evelyn Jo Johnson/McGraw Hill; (microsporangia): Karl J. Niklas; (d): TravelMuse/Alamy Stock Photo



c. *Ephedra*, a type of gnetophyte



d. *Welwitschia*, a type of gnetophyte

40 kg (Fig. 23.19*a*). Cycads exhibit the life cycle of a gymnosperm, except they are pollinated by insects rather than by wind. Also, the pollen tube bursts in the vicinity of the archegonium, and multiflagellated sperm swim to reach an egg.

Cycads were plentiful in the Mesozoic era at the time of the dinosaurs, and it's likely that dinosaurs fed on cycad seeds. Now, cycads are in danger of extinction, because they grow very slowly.

Ginkgoes

Although **ginkgoes** are plentiful in the fossil record, they are represented today by only one surviving species of tree, the *Ginkgo biloba*. Ginkgoes are **dioecious,** which means that a single plant produces either male or female reproductive structures, but not both (Fig. 23.19*b*). The fleshy seeds, which ripen in the fall, give off such a foul odor that male trees are usually preferred for planting. Ginkgo trees are resistant to pollution and do well along city streets and in city parks. Ginkgo is native to China, and in Asia, ginkgo seeds are considered a delicacy. Extracts from ginkgo trees have been used to improve blood circulation.

Like cycads, the pollen tube of ginkgo bursts to release multiflagellated sperm that swim to the egg produced by the female gametophyte, located within an ovule.

Gnetophytes

Gnetophytes are represented by three living genera and 95 species of plants that are very diverse in appearance. In all gnetophytes, xylem is structured similarly, none have archegonia, and their strobili (cones) have a similar construction. The reproductive structures of some gnetophyte species produce nectar, and insects play a role in the pollination of these species.

Gnetum, which occurs in the tropics, consists of trees or climbing vines with broad, leathery leaves arranged in pairs. *Ephedra,* occurring only in southwestern North America and Southeast Asia, is a shrub with small, scalelike leaves (Fig. 23.19*c*). Ephedrine, a medicine with serious side effects, is extracted from *Ephedra. Welwitschia,* living in the deserts of southwestern Africa, has only two enormous, straplike leaves that fray as it gets older (Fig. 23.19*d*).

Angiosperms

Angiosperms (Gk. *angion,* "vessel"; *sperma,* "seed") are the flowering plants. They are an exceptionally large and successful group, with roughly 370,000 species thought to exist—six times the number of all other plant groups combined. Angiosperms live in all sorts of habitats, from fresh water to desert and from the frigid north to the torrid tropics. They range in size from the tiny, almost microscopic duckweed to *Eucalyptus* trees over 100 m tall.

It would be impossible to exaggerate the importance of angiosperms in our everyday lives. Angiosperms include all the hardwood trees of temperate deciduous forests and all the broadleaved evergreen trees of tropical forests. Also, all herbaceous (nonwoody

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THEME Evolution

The Evolutionary History of Maize

Approximately 12,000 years ago, humans began shifting away from a hunter-gatherer way of life and moved toward an agriculturally based one. As early groups of humans began to spend more time in the same location, and sedentary populations grew, they required larger and more readily available food sources. Over time, these early farmers increasingly selected plants that produced greater yields of food with less of an investment in time and energy.

Maize, which we commonly call corn, was first cultivated in Central America nearly 10,000 years ago in the highlands of Mexico. By the time Europeans were exploring Central America in the 1500s, over 300 varieties were being cultivated (Fig. 23B). All of these varieties can trace their ancestral lineage back to a wild Mexican grass known as teosinte.

Among the earliest structural changes that occurred during corn domestication was the loss of *shattering,* the process by which ears of wild corn break apart and disperse their grains (Fig. 23C). A mutation likely caused the ears of some corn plants to remain intact, a trait that is disadvantageous in nature but beneficial to humans. Nonshattering ears are much easier for humans to harvest.



Figure 23B Varieties of corn. Corn can be cultivated to have a specific color, texture, and sugar content. alonsoleon9/iStock/Getty Images

How did our ancestors artificially select for desirable traits? Hunter-gatherers often followed seasonal migration routes between base camps, returning to the same locations year after year. The disturbance of the natural vegetation at these sites would have provided an ideal site for the colonization of species that were being cultivated by these groups. If next season's seeds were collected from the plants that had the most desirable traits, then over time the frequency of plants with these traits would increase. A modern example, showing how quickly artificial selection can change corn, was conducted in 1896 by agricultural scientists selecting for high oil content in corn kernels. Only the top 20% of oil-producing individuals of each generation were allowed to reproduce, and after 90 generations, the average oil content of the corn kernels had increased 450%.

Harvesting plants with specific traits, however, can decrease the genetic diversity within the population. This decline, in turn, produces a genetic bottleneck, in which various traits are lost, while others increase in frequency, giving rise to a change in the phenotypic frequency within the population (see Section 16.1).

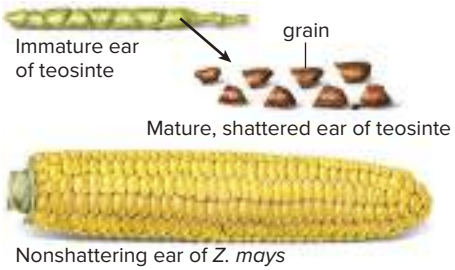


Figure 23C Ears and grain of modern corn and its ancestor, teosinte. Domesticated corn ears are larger than those of the ancestral grass teosinte. In addition, corn fruits are nonshattering, softer, and more edible than are grains of teosinte.

It was the practice of creating corn hybrids in the 1920s that significantly increased yields, as well as the genetic diversity of cultivars. Modern-day corn produces larger grain, generally a more robust plant, an increased amount of apical growth, and fruit with a higher protein and sugar content. In fact, corn has changed so much through artificial selection that domesticated corn can no longer successfully reproduce without human intervention.

Currently, the world's growing population, the loss of farmland, and climate change are straining global food production. The production of corn must now rely on molecular breeding. One method being used to create new strains employs CRISPR/Cas technology that is unique to corn, called the "CRISPR pollen method." This method uses CRISPR-edited genetically modified pollen to carry new versions of a gene to another corn plant's cells. For corn, this method is more effective and versatile than traditional methods of gene modification.

Information gained from genomic studies can also be used to customize corn varieties that increase yields under the growing environmental pressures of drought, pests, and microbial pathogens. Genetic engineering has opened up new windows of evolutionary options for the future of maize and, with that, the future of the humans who rely on this plant.

Questions to Consider

1. What potentially beneficial traits may have been lost during artificial selection in maize?
2. What other crop plants might have undergone a similar form of natural selection?

plants, such as grasses) and most garden plants are flowering plants. This means that all fruits, vegetables, nuts, herbs, and grains, which are the staples of the human diet, are angiosperms. They provide us with clothing, food, medicines, and other commercially valuable products. Over the past 12,000 years, humans have also artificially selected plants to serve us better, resulting in the cultivation of plants as food crops and for other uses. The Evolution feature, "The Evolutionary History of Maize," describes corn (*Zea mays*) as an example.

The flowering plants are called angiosperms because their ovules, unlike those of gymnosperms, are always enclosed within diploid tissues. In the Greek derivation of their name, *angio* ("vessel") refers to the ovary, which develops into a fruit, a unique angiosperm feature.

Origin and Radiation of Angiosperms

In general, the fossil record of plants is incomplete, because unlike bones and teeth, the soft parts of plants are often eaten or decayed before they can be preserved. This has created a vigorous debate among paleobotanists on the origins of flowering plants. One of the earliest known angiosperm fossils is